



Discrete Probability Distributions

• Mark Scheme

Further Statistics 1 Chapter 1: Discrete Probability Distributions. M = method mark A = accuracy mark (depends on M)
B = independent mark ft = follow through

Section A – Expected value

- 1 $E(X) = 0.1 + 0.6 + 1.2 + 0.8$ M1; = 2.7 A1.
- 2 $0.4 + 3k = 1 \Rightarrow k = 0.2$ B1; $E(X) = 0(0.2) + 1(0.2) + 2(0.4) + 3(0.2)$ M1; = 1.6 A1.
- 3 $\frac{1+2+3+4+5}{15} = \frac{15}{15} = 1$ B1; $E(X) = \sum x \cdot \frac{x}{15} = \frac{1+4+9+16+25}{15}$ M1; = $\frac{55}{15} = \frac{11}{3}$ A1.
- 4 $a + b + 0.5 = 1 \Rightarrow a + b = 0.5$ M1; $a + 2b + 0.9 + 0.8 = 2.3 \Rightarrow a + 2b = 0.6$ M1; $b = 0.1$ A1; $a = 0.4$ A1.
- 5(a) $k(4 + 3 + 2 + 1) + k = 1$ M1; $11k = 1 \Rightarrow k = \frac{1}{11}$ A1.
- 5(b) $E(X) = \frac{1}{11} (1 \times 4 + 2 \times 3 + 3 \times 2 + 4 \times 1 + 5 \times 1)$ M1; = $\frac{25}{11}$ A1.
- 5(c) $\frac{25}{11} = 2.27$, so need $P(X \geq 3)$ M1; = $\frac{2+1+1}{11} = \frac{4}{11}$ A1.
- 5(d) (1, 2) or (2, 1): $2 \times \frac{4}{11} \times \frac{3}{11}$ M1; = $\frac{24}{121}$ A1.

Section B – Variance

- 6 $E(X^2) = 0.5 + 4(0.25) = 1.5$ M1; $\text{Var}(X) = 1.5 - 1^2 = 0.5$ A1.
- 7 $E(X^2) = 0.1 + 1.2 + 3.6 + 3.2 = 8.1$ B1; $\text{Var}(X) = 8.1 - 2.7^2 = 0.81$ M1; standard deviation = 0.9 A1.
- 8 $E(X) = \frac{21}{6} = 3.5$ B1; $E(X^2) = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}$ M1; $\text{Var}(X) = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$ A1.
- 9 $E(X) = \frac{1+4+9+16}{10}$ M1; = 3 A1; $E(X^2) = \frac{1+8+27+64}{10} = 10$ M1; $\text{Var}(X) = 10 - 9 = 1$ A1.
- 10(a) $p + q = 0.5$ M1; $0.2 + 2p + 3q + 1.2 = 2.7 \Rightarrow 2p + 3q = 1.3$ M1; $q = 0.3$ A1; $p = 0.2$ A1.
- 10(b) $E(X^2) = 0.2 + 0.8 + 2.7 + 4.8 = 8.5$ M1 A1; $\text{Var}(X) = 8.5 - 2.7^2 = 1.21$ A1.
- 10(c) $\sigma = 1.1$, so $1.6 < X < 3.8$, i.e. $X = 2$ or 3 M1; probability = $0.2 + 0.3 = 0.5$ A1.

Section C – Functions of a random variable

- 11 $E(2X + 5) = 13$ B1; $\text{Var}(2X + 5) = 4 \times 3 = 12$ B1.
- 12 (a) $3 - 2(2) = -1$ B1; (b) $(-2)^2 \times 1.5 = 6$ B1; (c) $E(X^2) = \text{Var}(X) + [E(X)]^2 = 5.5$ B1.
- 13 $E(X^2) = 4.9$ and $E(X) = 2.1$ (or Y takes values $-1, 0, 3$) M1; $E(Y) = 4.9 - 2(2.1)$ M1; = 0.7 A1.
- 14 $2a^2 = 18$ M1; $a = 3$ (since $a > 0$) A1; $3a + b = 10$ M1; $b = 1$ A1.
- 15(a) $E(X) = 0.4 + 0.6 + 0.6 + 0.4$ M1; = 2 A1.
- 15(b) $E(X^2) = 0.4 + 1.2 + 1.8 + 1.6$ M1; = 5 A1; $\text{Var}(X) = 5 - 2^2 = 1$ A1.
- 15(c) $E(5 - 2X) = 1$ B1; $\text{Var}(5 - 2X) = 4$ B1.
- 15(d) $2(0.4) + 4(0.3) + 8(0.2) + 16(0.1)$ M1; = 5.2 A1.

Section D – Constructing distributions

- 16 $P(X = 0) = P(X = 3) = \frac{1}{8}$, $P(X = 1) = P(X = 2) = \frac{3}{8}$ M1 A1.
- 17 36 equally likely outcomes; $X = x$ in $2x - 1$ of them M1; $P(X = 1) = \frac{1}{36}$ B1; $P(X = x) = \frac{2x-1}{36}$: $\frac{1}{36}, \frac{3}{36}, \frac{5}{36}, \frac{7}{36}, \frac{9}{36}, \frac{11}{36}$ for $x = 1, \dots, 6$ A1.
- 18 $P(X = 0) = \frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$, $P(X = 2) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$ M1; $P(X = 1) = \frac{6}{10}$ A1; $E(X) = \frac{6+6}{10} = 1.2$ B1.



19 Profit values 4, 1, -2 *M1*; with probabilities $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}$ *A1*; $E(X) = \frac{4}{6} + \frac{1}{3} - 1$ *M1*; $= 0$, so the game is fair *A1*.

20(a) $P(X = 1) = 0.4$ *B1*; $P(X = 2) = 0.6 \times 0.4 = 0.24$ *B1*; $P(X = 3) = 0.6^2 = 0.36$ *B1*.

20(b) $E(X) = 0.4 + 0.48 + 1.08$ *M1*; $= 1.96$ *A1*.

20(c) $E(X^2) = 0.4 + 0.96 + 3.24 = 4.6$ *M1*; $\text{Var}(X) = 4.6 - 1.96^2$ *M1*; $= 0.7584$ *A1*.

20(d) $E(9 - 3X) = 9 - 3(1.96)$ *M1*; fair entry fee $= £3.12$ *A1*.

Roar-level challenge

21 Let $P(X = 0) = a$, $P(X = 1) = b$, $P(X = 2) = c$; $a + b + c = 1$ *M1*; $b + 2c = 1$ *M1*; $E(X^2) = 0.6 + 1^2 = 1.6$, so $b + 4c = 1.6$ *M1*; $c = 0.3$ *A1*; $b = 0.4$, $a = 0.3$ *A1*.